

Choosing the Right Evidence for Cybersecurity AI

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Thesis A checker that recognizes when retrieved security facts actually apply could improve cybersecurity AI while preserving the benefits of useful evidence.

Problem and impact Security AI can be misled by apparently relevant evidence, potentially wasting analysts' time and leading to poor security decisions (Hamzić et al., 2026; Liu et al., 2026).

Hypothesis A checker trained to predict whether evidence will help will improve accuracy on unfamiliar security questions, limit harm from mismatched facts, and outperform simpler relevance checks.

Primary base paper and gap CoRM-RAG (Liu et al., 2026) already learns whether evidence is useful. Our target is a checker that recognizes when security facts apply to unfamiliar questions and beats simpler checks while using facts equally often. A new contribution has not yet been demonstrated. Hamzić et al. (2026) provide the broader CTI foundation.

Dataset and example Development used 2,500 CTIBench questions; external testing used 1,247 qualifying SecEval questions from 10 security source families, with retrieved MITRE ATT&CK 19.1 facts (Li et al., 2023). SecEval answers were AI-generated/calibrated; independent human review is pending.

Real example, shortened: Which tool best tests a web application for SQL injection? A: sqlmap; B: wget; C: Find Security Bugs; D: OWASP O-Saft. Released answer: A.

Experiment in simple terms Freeze the checker before testing. Have Llama 3.1 8B and Qwen 2.5 7B answer every question with and without six retrieved facts, producing 4,988 fresh answers. Let the checker choose which answer to use; compare with always using facts, never using them, and four simpler checkers, including comparisons using facts equally often.

Results The external test did not support the hypothesis.

Model	No extra facts	Always add facts	Our checker
Llama 3.1 8B	86.53%	84.20%	86.21%
Qwen 2.5 7B	87.09%	85.24%	87.25%

The checker used facts on 6.82% of questions; its average change from no extra facts was -0.08 percentage points (95% interval: -0.52 to +0.36). The planned improvement and comparison tests failed. Earlier pilot gains of +12.24 and +9.64 points did not transfer to this test.

APA REFERENCES

Hamzić, D., Skopik, F., Landauer, M., Wurzenberger, M., & Rauber, A. (2026). *Beyond RAG for cyber threat intelligence: A systematic evaluation of graph-based and agentic retrieval*. arXiv. <https://doi.org/10.48550/arXiv.2604.11419>

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